

Problem 2-9

A company produces two products, A and B. The sales for A is at least 80% of the total sales of both A and B. However, the company cannot sell more than 110 units of A per day. Both products use one raw material, of which the maximum daily availability is 300 kg. The usage rates of the raw material are 2 kg per unit of A, and 4 kg per unit of B. The profit units for A and B are \$40 and \$90, respectively. Determine the optimal product mix for the company.

Diet Problem Example

Usman uses at least 800 lb of special feed daily. The special feed is a mixture of corn and soybean meal with the following compositions:

Feedstuff	Protein (lb/lb)	Fiber (lb/lb)	Cost (\$/lb)
Corn	0.09	0.02	0.30
Soybean meal	0.60	0.06	0.90

Dietary requirements of the special feed are at least 30% protein and at most 5% fiber.

Then the goal is to determine the daily minimum-cost feed mix.

Problem 2-8

A company that operates 10 hours a day manufactures two products on three sequential processes. The following table summarizes the data:

Product	Process 1	Process 2	Process 3	Unit Profit
1	10	6	8	\$20
2	5	20	10	\$30

Problem 2-13

Chemistry Lab uses raw materials I and II to produce two domestic cleaning solutions, A and B. The daily availability's of raw materials I and II are 150 and 145 units, respectively. One unit of solution A consumes 0.5 units of raw material I and 0.6 units of raw material II. One unit of solution B uses 0.5 units of raw material I and 0.4 units of raw material II. The profits per unit of solutions A and B are \$8 and \$10, respectively. The daily demand for solution A lies between 30 and 150 units, and that for solution B between 40 and 200 units. Find the optimal production amounts of A and B.

Example

A person requires at least 10, 12 & 12 units chemicals A, B & C respectively for his garden. A liquid product contains 5, 2 & 3 units of A, B & C respectively, per jar. A dry product contains 3, 3 and 4 units of A, B and C respectively per carton. If the liquid product cost is Rs. 3 per jar & dry product cost is Rs. 2 per carton. How many of each should be purchased to minimize the cost and meet the requirement. Only formulate the above problem as standard L.P.P.

Problem 2-19

Top Toys is planning a new radio and TV advertising campaign. A radio commercial costs \$300 and a TV ad costs \$2000. A total budget of \$20,000 is allocated to the campaign. However, to ensure that each medium will have at least one radio commercial and one TV ad, the most that can be allocated to either medium cannot exceed 80% of the total budget. It is estimated that the first radio commercial will reach 5000 people, with each additional commercial reaching only 2000 new ones. For TV, the first ad will reach 4500 people, and for each additional ad, an additional 3000. How should the budgeted amount be allocated between radio and TV?

Problem 2-31

John must work at least 20 hours a week to supplement his income while attending school. He has the opportunity to work in two retail stores. In store 1, he can work between 4.5 and 12 hours a week, and in store 2, he is allowed between 5.5 and 10 hours. Both Stores pay the same hourly wage. In deciding how many hours to work in each store, John wants to base his decision on work stress. Based on interviews with present employees, John estimates that, on an ascending scale of 1 to 10, the stress factors are 8 and 6 at stores 1 and 2, respectively. Because stress mounts by the hour, he assumes that the total stress for each store at the end of the week is proportional to the number of hours he works in the store. How many hours should John work in each store?

Problem 2-40

Fox Enterprises is considering six projects for possible construction over the next four years. Fox can undertake any of the projects, partially or completely. A partial undertaking of a project will prorate both the return and cash outlays proportionately. The expected (present value) returns and cash outlays for the projects are given in the following table.

Project	Cash outlay (\$1000)				Return (\$1000)
	Year 1	Year 2	Year 3	Year 4	
1	10.5	14.4	2.2	2.4	324.00
2	8.3	12.6	9.5	3.1	358.00
3	10.2	14.2	5.6	4.2	177.50
4	7.2	10.5	7.5	5.0	148.00
5	12.3	10.1	8.3	6.3	182.00
6	9.2	7.8	6.9	5.1	123.50
Available funds (\$1000)	60.0	70.0	35.0	20.0	

- Formulate the problem as a linear program, and determine the optimal project mix that maximizes the total return using AMPL, Solver, or TORA. Ignore the time value of money.
- Suppose that if a portion of project 2 is undertaken, then at least an equal portion of project 6 must be undertaken. Modify the formulation of the model, and find the new optimal solution.
- In the original model, suppose that any funds left at the end of a year are used in the next year. Find the new optimal solution, and determine how much each year “borrows” from the preceding year. For simplicity, ignore the time value of money.
- Suppose in the original model the yearly funds available for any year can be exceeded, if necessary, by borrowing from other financial activities within the company. Ignoring the time value of money, reformulate the LP model, and find the optimum solution. Would the new solution require borrowing in any year? If so, what is the rate of return on borrowed money?

Quiz

A firm manufactures pain-relieving pills in two sizes, A and B. Size A contains 4 grains of element X, 7 grains of element Y, and 2 grains of element Z. Size B contain 2 grains of element X, 10 grains of element Y, and 8 grains of element Z. It is found by users that it requires at least 12 grains of element X, 34 grains of element Y, and 24 grains of element Z to provide immediate relief. It is required to determine the least number of pills a patient should take to get immediate relief. Only formulate the above problem as standard L.P.P

Problem 2-34

An industrial recycling center uses two scrap of metals, A and B, to produce a special alloy. Scrap A contains 6% aluminum, 3% silicon, and 4% carbon. Scrap B has 3% aluminum, 6% silicon, and 3% carbon. The costs per ton for scraps A and B are \$100 and \$80, respectively. The specifications of the special alloy require that (1) the aluminum content must be at least 3% and at most 6%, (2) the silicon content must be between 3% and 5%, and (3) the carbon content must be between 3% and 7%. Determine the optimum mix of the scraps that should be used in producing 1000 tons of the alloy.

Problem 2-10

Alumco manufactures aluminum sheets and aluminum bars. The maximum production capacity is estimated at either 800 sheets or 600 bars per day. The maximum daily demand is 550 sheets and 560 bars. The profit per ton is \$40 per sheet and \$35 per bar. Determine the optimal daily production mix.

Problem 3-9

JoShop manufactures three products whose unit profits are \$2, \$5, and \$3, respectively. The company has budgeted 80 hrs of labor time and 65 hrs of machine time for the production of the three products. The labor requirements per unit of products 1, 2, and 3 are 2, 1, and 2 hrs, respectively. The corresponding machine-time requirements per unit are 1, 1, and 2 hrs. JoShop regards the budgeted labor and machine hours as goals that may be exceeded, if necessary, but at the additional cost of \$15 per labor hour and \$10 per machine hour. Formulate the problem as an LP, and determine its optimum solution using TORA, Solver, or AMPL